COMPUATIONAL APPROACHES TO MIGRATION RESEARCH

# Computational Approaches to Migration Research

Institute of Sociology and Demography (ISD), University of Rostock, Germany

## Instructor Info

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## Course Info

Pre-requirement: None

Thursdays

9:00-11:00

PC pool 226, ISD

## Lab Info

Every other week

9:00-11:00

PC pool 226, ISD

## Overview

Migration, as an area of research, has attracted increased attention from different fields of science. There has been an increase in the number of scientific publications focused on migration, even though the share of migrants worldwide, relative to the population size, has stayed largely stable. Migration is, however, notoriously difficult to measure. There has been much emphasis on how to address data shortages in this area of research. With the increased digitization and the advent of social media, big data, and digital trace data, one of the widely used approaches in migration research is to repurpose data. Since such repurposed data has not been produced for research, researchers must make several decisions and overcome challenges to use it to answer migration research questions.

Computational approaches to migration research is a course designed to equip students with knowledge about migration theories, concepts, their operationalization and measurement, and methodological aspects of repurposing various sources of data from social media, online networks, geospatial data, metadata of scientific publications, and similar sources for migration research.

## Learning Objectives

By the end of this course, students will be familiar with migration concepts and theories, different sources of data, and computational approaches to migration research. They will have knowledge of different research questions, data sources, and possible answers for the questions provided in Computational Social Science. Through hands-on lab sessions, they will acquire or advance their technical skills in R, Python, and SQL to use large-scale data. They will also learn about ethics in digital and computational migration research.

## Reading Materials

Every week's materials include required and recommended readings specific to that week's topic, and some of the main ones are listed below.

### Required Text


Drouhot, L. G., Deutschmann, E., Zuccotti, C. V., & Zagheni, E. (2022). Computational approaches to migration and integration research: Promises and challenges. *Journal of Ethnic and Migration Studies*, *0*(0), 1–19. https://doi.org/10.1080/1369183X.2022.2100542


### Recommended Texts


Akbaritabar, A., Dańko, M. J., Zhao, X., & Zagheni, E. (2025). Global subnational estimates of migration of scientists reveal large disparities in internal and international flows. *Proceedings of the National Academy of Sciences*, *122*(15), e2424521122. https://doi.org/10.1073/pnas.2424521122

Alessandretti, L., Aslak, U., & Lehmann, S. (2020). The scales of human mobility. *Nature*, *587*(7834), 402–407. https://doi.org/10.1038/s41586-020-2909-1

de Haas, H. (2021). A theory of migration: The aspirations-capabilities framework. *Comparative Migration Studies*, *9*(1), 8. https://doi.org/10.1186/s40878-020-00210-4

Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2022, April). *Digital and Computational Demography* (tech. rep.). SocArXiv. https://doi.org/10.31235/osf.io/7bvpt

Massey, D. S., Arango, J., Hugo, G., Kouaouci, A., Pellegrino, A., & Taylor, J. E. (1993). Theories of International Migration: A Review and Appraisal. *Population and Development Review*, *19*(3), 431. https://doi.org/10.2307/2938462

Salah, A. A., Bircan, T., & Korkmaz, E. E. (2022). New data sources and computational approaches on migration and human mobility. In *Data Science for Migration and Mobility Studies*. British Academy.

Tjaden, J. (2021). Measuring migration 2.0: A review of digital data sources. *Comparative Migration Studies*, *9*(1), 59. https://doi.org/10.1186/s40878-021-00273-x

# FAQs

## How to access the course materials?

There is a private GitHub repository where all students are added in the first session. Every week's slides, hands-on materials, code, and data will be uploaded there.

## What to do at home before/after sessions?

1) Study the reading materials and prepare some questions or discussion points. 2) Run codes in hands-on materials, reproduce the examples, extend them, and note the unclear points.

## Where to learn basics of R, Python, SQL?

The course materials start from the basics. Start by practicing with the provided examples. Materials also provide links to other introductory courses.

## Can we use Artificial Intelligence to help us?

AI is a tool similar to any other tool out there. You should be "in control", and take "responsibility" for using it. Follow the ISD institute's reporting protocol to share how you used it.

## Where to learn about academic writing?

1) https://writingcenter.gmu.edu/writing-resources/imrad/writing-an-imrad-report
2) https://www.nature.com/scitable/topicpage/effective-writing-13815989/

## Grading Scheme

The final grade will be on a 1-6 scale with these meanings: 1.0: "sehr gut"/ Excellent; 1.3: "sehr gut"/ Excellent; 1.7: "gut"/ Good; 2.0: "gut"/ Good; 2.3: "gut"/ Good; 2.7: "befriedigend"/ Satisfactory; 3.0: "befriedigend"/ Satisfactory; 3.3: "befriedigend"/ Satisfactory; 3.7: "ausreichend"/ Pass; 4.0: "ausreichend"/ Pass.

Active participation in course discussions during the semester will inform my evaluation of your understanding of the topics. If you have specific concerns that might inhibit your active participation in class discussions of the reading materials, please inform me at the start of the course.

## Course Examination

The mode of examination is a "term paper" ("Hausarbeit" in German). In the penultimate week, I will share a list of topics (from the course schedule below) and ask you to select one. You will use the reading list for that topic, search for other literature, plus use the data sources and discussed research questions, and write an essay that should include these sections: a) introduction, b) background and literature, c) methods and analytical strategy, d) results, e) discussion and conclusion (including possible limitations of your method and results), f) references (to the literature, data sources, etc., that you used).

## Diversity and Inclusivity Statement

To become a great social scientist, a student needs to learn to add a "why" before every statement. The same statements that others hear without questioning. Students need to develop sharp observation skills and find a new angle in seeing familiar social phenomena. This is how they can observe inequalities and previously less explored aspects of society. This is the same society in which we all live, however, those with sharpened observation skills can identify inequalities in a familiar context. They can question the ordinary, find reasons behind observed phenomena, and imagine counterfactual scenarios for the future. This practice helps students develop critical thinking skills. I converse with students, write their ideas on the whiteboard, and ask them to explain those ideas. I create a safe space where every student feels welcome to contribute, which helps students develop their own ideas confidently.

## Students with Special Needs

Please inform me as soon as possible of special needs that you may have. The sooner you notify me, the better I will be able to coordinate with the institute and other responsible persons to accommodate your special needs.

## Students' Mental Health and Well-being

Academic life and university studies can be very stressful and challenging. Please inform me or seek proper professional help if you are concerned with your mental health and well-being. Seeking help is one of the most courageous things to do.

## Integrity, Plagiarism, and Ethics of Research

Students are expected to prepare their final assignments independently. Any violation of this will not be tolerated and will be appropriately reported to the faculty and university according to the institution's guidelines. If you use a resource in your work, cite it properly and give due credit by following established guidelines, e.g., APA: https://apastyle.apa.org/style-grammar-guidelines/citations/plagiarism.

## Credits

- The logo of the course is designed using Google Gemini.
- The syllabus LaTeX template is modified based on https://www.overleaf.com/latex/templates/inzane-syllabus-template/zpgzsdfxvqmz.

## Class and Lab Schedule

### PART 1: Introductory topics

| | | |
|---|---|---|
| Week 1 | Introduction | What is "Computational Approaches to Migration Research"? Who is an (e/im)migrant? |
| | Accessing course materials | Importance of reproducibility; Version control, Git, GitHub |

### PART 2: Migration theories

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|---|---|---|
| Week 2 | Functionalist | Neo-classical; Equilibrium; Push-Pull; Balanced growth |
| Week 3 | Historical-structuralist | Neo-Marxist; Conflict theory; Asymmetrical growth; Dependency; World Systems theory |
| Week 4 | Aspirations/capabilities | The importance of agency; Attraction versus repulsion; Mobility bias; (In)voluntary (im)mobility; Integration |
| | | Migration networks; Transnationalism; Remittances; Circular/Cumulative causation and self-sustaining feature of migration |

### PART 3: Methods of migration research

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|---|---|---|
| Week 5 | Methods | IMISCOE Taxonomy; Quantitative/computational; Qualitative; Mixed methods |
| | Units of analysis, and dichotomies | Internal/International; Individual/Household; Temporary/Permanent; Planned/Flight (asylum seekers; refugees; forced migration); Illegal/Legal |
| Week 6 | (Under)representation | (Under)represented regions/countries in migration research; Reasons behind under-exploration; Gender-blindness of migration research |
| | Other methodological aspects | Operationalization of migration theories and concepts; Measurement issues and limitations |

### PART 4: Data sources and applications

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|---|---|---|
| Week 7 | Traditional | Census (IPUMS International); Survey (SHARE, SOEP); Register |
| | | Re-purposing data for migration research |
| Week 8 | Digital traces | Facebook; Twitter; BluSky; Other social media; Online news |
| Week 9 | Digital traces | LinkedIn; Online search/Google Trends; Email and IP; Wikipedia data; Air travel data |
| Week 10 | Digital traces | Geo-spatial data; Mobile phone; Sensors/wearables; Satellite data |
| Week 11 | Digital traces | Bibliometric data; Patents; Online CVs; Synthetic and simulated data |

### PART 5: Conclusions

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|---|---|---|
| Week 12 | Ethics | Ethical considerations in computational approaches to migration research; Informed consent; Policy debates vs. real-world migration |
| Week 13 | Concluding discussions | Limitations; Pitfalls of digital trace data and computational approaches; Representativeness; Thick vs Big data; Mixed Methods |
| | For final examination | LIST OF FINAL ASSIGNMENTS |
| Week 14 | Discussing final examination | Share your selected topic; 6 weeks to prepare a short essay and empirical analysis on one of the topics |